



DO-IT-YOURSELF: PROJECT

EFY Note

The source code of this project is available for free download at source.efymag.com

DHT_sensor_library-1.4.2 and DHT sensor library version 1.4.3. During testing it was found that without these libraries the code could not be compiled.

Construction and testing

The circuit can be assembled on a breadboard or general-purpose PCB. The author's prototype is shown in

PARTS LIST

Semiconductors:

Board1	- Arduino Uno board
BT1	- HC-05 Bluetooth module
SM1	- Soil moisture sensor
SM2	- DHT11 temperature and humidity sensor
LDR1	- 10-kilo-ohm LDR
IC1	- L293D motor driver

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 220-ohm
R2	- 10-kilo-ohm

Miscellaneous:

M1, M2	- 12V DC motor
M3	- Servo motor (Tower Pro)
BATT.1	- 12V Battery
	- Robot chassis
	- Wheels for the robot
	- Jumper wires

Fig. 3. On connecting the circuit to a 12V DC supply, the onboard LED on Bluetooth HC-05 will start blinking at a fast rate. When you pair it successfully with your Bluetooth app in your mobile phone, the LED will blink at a slower rate (two blinks per second). Open the Bluetooth

Terminal HC-05 app from the mobile phone, select HC-05 again.

If everything's fine, you can enter control commands (F,B,R,L) in the app to move the robot. On entering 'S' command, the robot will stop and all the sensor data can be seen on your mobile phone, as shown in Fig. 4.

Soil moisture sensor SM1 should be properly fixed to the servo arm/horn using either glue gun or screws. When servo motor pulls out soil moisture sensor from the soil, you can see the moisture level drastically reduces to 5, as shown in Fig. 5, indicating that the sensor is out of the soil. **EFY**



Ashutosh M. Bhatt is M.Tech in embedded systems. Currently, he is lecturer of electronics and radio engineering at Government Polytechnic, Jamnagar, Gujarat

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